

DAKSH PANCHAL

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EDUCATION

Delhi Technological University Aug 2024 – May 2028
B.Tech Computer Science and Engineering | GPA: 8.87/10 (up to 3rd sem) Delhi
JEE Main 2024: AIR 11,214 (99.3 Percentile) 2024
Goodley Public School: CBSE XII (95.4 Percentage) 2024

PERSONAL PROJECTS

Mars Exploration Drone Software Stack | GitHub 2026

- Designed a Pixhawk–Raspberry Pi flight software stack with MAVLink-based dual-pilot control (<100ms latency), telemetry routing at 50Hz, and failsafe manual override; stress-tested over 50+ flight hours with no critical failures
- Implemented an onboard processing pipeline combining 64MP imaging with a 3-sensor atmospheric suite (SHT45, BME688, particulate sensor) and dual-RF telemetry for redundant, fault-tolerant communication

Roblog – AI-Powered SEO Blog Generation Platform | GitHub 2026

- Developed a containerized full-stack SaaS (FastAPI + Next.js + MongoDB) automating SEO-optimized blog generation, cutting manual content creation time by ~80% via a structured LLM inference and prompt engineering pipeline
- Exposed scalable REST APIs for full CRUD operations; deployed via Docker Compose with CI/CD pipeline, reducing deployment overhead by 40%

Chess Engine – Game AI System | GitHub 2025

- Engineered a C++ chess engine with alpha-beta pruned minimax search (depth 8, <200ms per move) leveraging efficient data structures (bitboards) for move generation across the full game tree
- Integrated a neural network evaluator trained on 500K+ positions, improving move quality ~18% over handcrafted heuristics; shipped a cross-platform Flutter UI with modular design for real-time engine interaction

EXPERIENCE – RESEARCH TEAM: UNMANNED GROUND VEHICLE

Mars Exploration Drone | ISDC 2026 | Repository | Report 2026

- Led MAVLink-based flight software integration on Pixhawk–Raspberry Pi hardware, achieving <100ms control latency and 50Hz telemetry; zero safety-critical failures across the full 50+ hour ISDC 2026 test campaign
- Integrated 64MP imaging with heterogeneous sensor fusion across temperature, gas, and particulate channels (SHT45, BME688); dual-RF telemetry achieved zero communication dropout across all pre-competition test flights

Autonomous Ground Vehicle | IGVC 2025 USA | Repository | Report 2025

- Engineered a ROS2-based autonomy stack with SLAM and DWA local planning, integrating multi-sensor fusion (LiDAR, stereo camera, IMU) to achieve 95% waypoint accuracy in dynamic outdoor environments

ACHIEVEMENTS

Top 10 Global Rank – ISDC 2026 | International Space Drone Challenge **Manipal, India | 2026**
Top 10 Global Rank – IGVC 2025 | Autonomous Navigation Challenge **Michigan, USA | 2025**

POSITION OF RESPONSIBILITY

UGV-DTU Research Team Aug 2024 – Present

- Software Head (2025–Present):** Lead a team of 15+ software engineers; own software architecture, codebase standards, technical direction, and Agile sprint planning across all competition deliverables
- Software Engineer (2024–2025):** Developed core autonomy, navigation, and sensor fusion modules; maintained Git-based version control workflows and code review processes across the team

TECHNICAL SKILLS

Languages: C++17/20, Python, Go, JavaScript/TypeScript
Frameworks & Libraries: FastAPI, Next.js, React, Node.js, OpenCV, TensorFlow
Databases: MongoDB, PostgreSQL
Systems & DevOps: Linux, Docker, Git, CI/CD, REST APIs, Microservices, System Design
Embedded & Robotics: ROS2, MAVLink, SLAM, Kalman/EKF, Gazebo, Arduino, UART/SPI/I2C